CMR INSTITUTE OF TECHNOLOGY

PROJECT TITLE: SMART HEALTH MONITORING DEVICE

TEAM NUMBER : 09

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SMART HEALTH MONITORING DEVICE

INTRODUCTION

“A Smart Health Monitoring Device “ is an IoT-based system used to monitor vital health parameters such as heart rate,body temperature, and oxygen level in real time.The device collects data using sensors and processes it through a microcontroller. This data can be accessed remotely by doctors or caregivers, helping in early detection of health problems and reducing the need for frequent hospital visits.

BRIEF HISTORY OF SMART HEALTH MONITORING DEVICE

Health monitoring initially relied on manual methods such as periodic checkups and basic medical instruments. With the advancement of electronics and embedded systems, portable medical devices like digital thermometers and heart rate monitors were developed. The introduction of microcontrollers and wireless communication led to remote health monitoring systems. In recent years, the integration of IoT, sensors and mobile applications has resulted in smart health monitoring devices that enable real time data tracking , remote access, and timely medical alerts.

PROBLEM STATEMENT

1. Traditional health monitoring requires frequent hospital visits, which is time consuming and inconvenient for patients.
2. Continuous monitoring of vital signs is difficult using manual or conventional medical methods.
3. Emergency health conditions may go unnoticed due to the lack of real time monitoring and alert systems.
4. Doctors and caregivers cannot monitor patients remotely in existing basic healthcare systems.
5. Elderly and chronic patients need constant supervision, which is not always possible.

DESIGN THINKING PROCESS

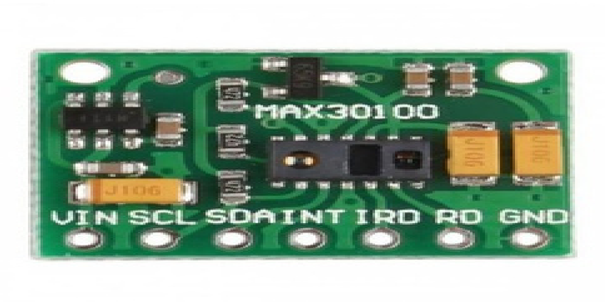
- **EMPATHIZE**: In today’s busy world people often ignore regular health checkups due to lack of time or access. Elderly patients , those with chronic illness and even fitness lovers need an easy way to track their vital signs daily. A smart health monitoring device helps by providing real time updates on wealth, reducing hospital visits giving peace of mind and caregivers .It’s a step towards proactive and accessible personal healthcare.
- **DEFINE**: Traditional health monitoring is manual and intermittent leading to incomplete data and delayed insights. This project addresses the need for a seamless, automated system that continuously tracks the vital signs (blood pressure, oxygen and heart rate) and securely saves data enabling proactive health management and better communication with health care providers.
- **IDEATE**: Wearable devices to track vital signs like heart rate ,SpO2 ,temperature
- Mobile app integration to show real time data and alerts.
- Cloud storage
- AI based health alerts
- **PROTOTYPE DISCRIPTION:**

1. ESP32: Microcontroller developed by **Espressif systems.** It is used in IoT and embedded systems.

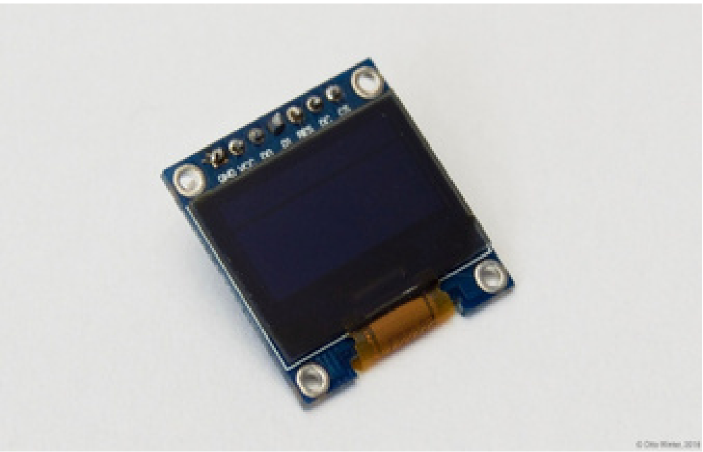
2. MAX30100: Integrated pulse oximetry and heart rate sensor.
3. OLED DISPLAY: It shows real time information and data clearly in text.

ESP32:

- **MAX30100**

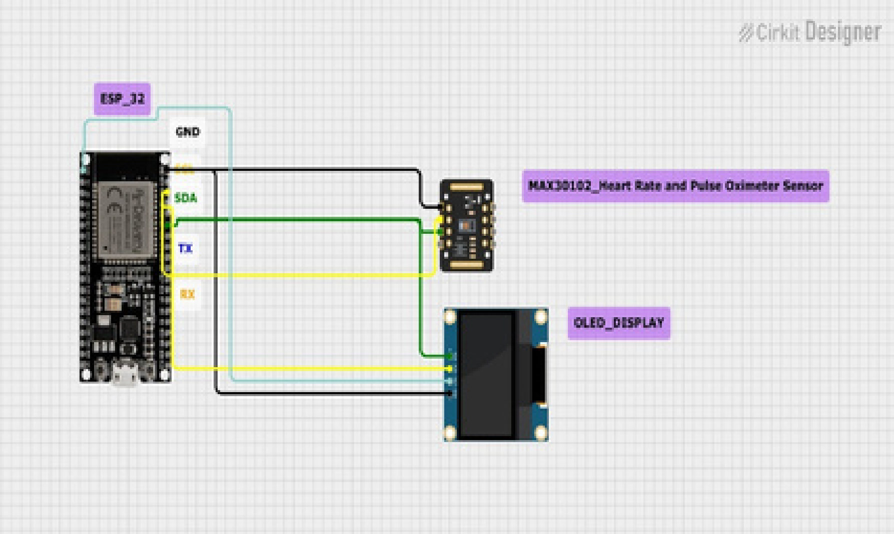


- OLED DISPLAY



- **TEST:** Technical validation:vital sign like heart rate,SpO2 and body temperature

CIRCUIT CONNECTION:



PROGRAM CODE:

```
#include <Wire.h>
#include "MAX30100_PulseOximeter.h"

// --- OLED Display Libraries ---
#include "OakOLED.h"

// --- ESP32 I2C Pin Definitions ---
#define I2C_SDA_PIN 21
#define I2C_SCL_PIN 22

// --- Project Constants ---
#define REPORTING_PERIOD_MS 1000

// 🚫 REMOVED: const uint8_t IR_50MA = 3;
// The manual definition is removed. We will use the library's enum constant directly.
```

```
// --- Global Objects ---

OakOLED oled; // Use your original OakOLED object

PulseOximeter pox; // MAX30100 sensor object


uint32_t tsLastReport = 0;


// Placeholder for the beat bitmap

const unsigned char bitmap [] PROGMEM =

{

0x00, 0x00, 0x00, 0x01, 0x80, 0x18, 0x00, 0x0f, 0xe0, 0x7f, 0x00, 0x3f, 0xf9, 0xff, 0xc0,

0x7f, 0xf9, 0xff, 0xc0, 0x7f, 0xff, 0xff, 0xe0, 0x7f, 0xff, 0xff, 0xe0, 0xff, 0xff, 0xff, 0xf0,

0xff, 0xf7, 0xff, 0xf0, 0xff, 0xe7, 0xff, 0xf0, 0xff, 0xe7, 0xff, 0xf0, 0xff, 0xf0, 0x7d, 0xdb, 0xff, 0xe0,

0x7f, 0x9b, 0xff, 0xe0, 0x00, 0x3b, 0xc0, 0x00, 0x3f, 0xf9, 0xff, 0xc0, 0x3f, 0xfd, 0xbf, 0xc0,

0x1f, 0xfd, 0xbf, 0x80, 0x0f, 0xfd, 0x7f, 0x00, 0x07, 0xfe, 0x7e, 0x00, 0x03, 0xfe, 0xfc, 0x00,

0x01, 0xff, 0xf8, 0x00, 0x00, 0xff, 0xf0, 0x00, 0x00, 0x7f, 0xe0, 0x00, 0x00, 0x3f, 0xc0,

0x00,

0x00, 0x0f, 0x00, 0x00, 0x00, 0x06, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00

};


// Callback fired when a pulse is detected

void onBeatDetected()

{

   // Lightly draw the beat indicator on the OLED

   oled.drawBitmap(60, 20, bitmap, 28, 28, 1);

   oled.display();

}


void setup()

{

   Serial.begin(115200);
```

```
// --- ESP32 I2C Initialization (Crucial!) ---
Wire.begin(I2C_SDA_PIN, I2C_SCL_PIN);

// --- OLED Initialization ---
oled.begin();
oled.clearDisplay();
oled.setTextSize(1);
oled.setTextColor(1);
oled.setCursor(0, 0);
oled.println("Starting Sensor...");
oled.display();

// --- MAX30100 Initialization ---
Serial.print("Initializing pulse oximeter...");
if (!pox.begin()) {
  Serial.println("FAILED (0x57 not found!)");
  oled.clearDisplay();
  oled.setCursor(0, 0);
  oled.println("MAX30100 FAILED");
  oled.display();
  for (;;); // Halt
} else {
  Serial.println("SUCCESS");
  oled.clearDisplay();
  oled.setCursor(0, 0);
  oled.println("SUCCESS");
  oled.display();
}
```

```
    // Register the beat detection callback
    pox.setOnBeatDetectedCallback(onBeatDetected);

    // Set IR LED current using the library's correct enum constant
    // NOTE: If this fails, try "MAX30100_LED_CURRENT_50MA" instead.
    pox.setIRLedCurrent(MAX30100_LED_CURR_50MA);
}

void loop()
{
    pox.update();

    if (millis() - tsLastReport > REPORTING_PERIOD_MS) {
        float heartRate = pox.getHeartRate();
        float spo2 = pox.getSpO2();

        // --- USB Serial Debug ---
        Serial.print("HR: ");
        Serial.print(heartRate, 2);
        Serial.print(", SpO2: ");
        Serial.println(spo2, 2);

        // --- OLED Display Update ---
        oled.clearDisplay();

        // Heart Rate Display
        oled.setTextSize(1);
        oled.setCursor(0, 0);
        oled.println("Heart BPM:");
        oled.setTextSize(2);
```

```
    oled.setCursor(0, 16);

    oled.println(heartRate);


    // SpO2 Display

    oled.setTextSize(1);

    oled.setCursor(0, 35);

    oled.println("SpO2 (%):");

    oled.setTextSize(2);

    oled.setCursor(0, 50);

    oled.println(spo2);


    oled.display();

    tsLastReport = millis();

  }

}
```

USER TESTING AND FEEDBACK:

Sample output:

```
HR: 29.21, SpO2: 0.00
HR: 0.00, SpO2: 0.00
HR: 0.00, SpO2: 0.00
HR: 0.00, SpO2: 0.00
HR: 0.00, SpO2: 0.00
HR: 0.00, SpO2: 0.00
HR: 14.91, SpO2: 0.00
HR: 14.91, SpO2: 0.00
HR: 0.00, SpO2: 0.00
```

Feedback:

- **It accurately measures health parameters like heart rate, SpO2.**
- **Useful, efficient and user friendly**

CONCLUSION:

The smart health monitoring device is an efficient and practical solution for real time health monitoring. It accurately measures vital signs such as heart rate, SpO2 and displays them clearly on OLED display . With ESP32 integration ,data can be transmitted wirelessly for remote monitoring. The device compact and user friendly making it suitable for personal health tracking and preventive healthcare.

FUTURE SCOPE:

- Integration with mobile applications for real time monitoring.
- Cloud storage for long term health data analysis.
- Development of a wearable and compact design.